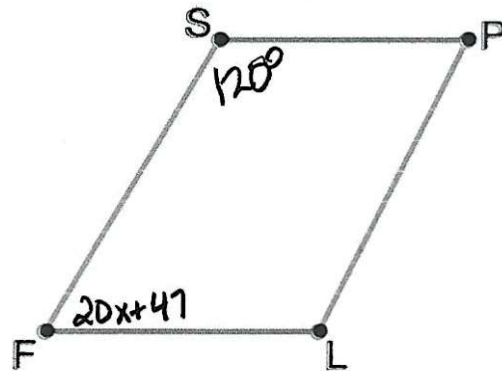


Unit 8: Worked Examples

1. In $SPLF$, $m\angle S = 120^\circ$, and $m\angle F = (20x + 47)^\circ$. If $x = \frac{1}{2}$, determine whether the figure is a parallelogram.

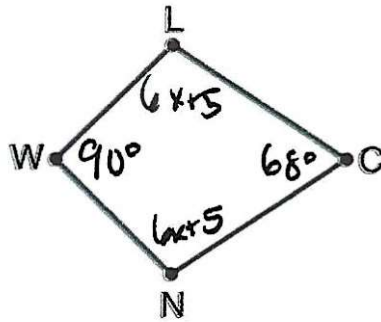


$$20\left(\frac{1}{2}\right) + 47 = 57^\circ$$

$$120 + 57 = 177^\circ$$

NOT a parallelogram.

2. Kite $WLCN$ is shown with $m\angle W = 90^\circ$, $m\angle C = 68^\circ$, and $m\angle N = (6x + 5)^\circ$. What is the value of x ?



$$12x + 10 + 90 + 68 = 360$$

$$12x + 168 = 360$$

$$12x = 192$$

$$x = 16$$